

Code No: 157DX**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year I Semester Examinations, July/August - 2022****SOFTWARE PROCESS AND PROJECT MANAGEMENT****(Common to CSE, IT)****Time: 3 Hours****Max.Marks:75**

Answer any five questions
All questions carry equal marks

1. Describe software maturity framework. [15]
2. What are process reference models? Explain them. [15]
3. Describe evolution of software economics. [15]
4. Explain engineering and production stages in detail. [15]
5. Explain software process workflow and iteration workflow. [15]
6. What are planning guidelines? What is pragmatic planning? Explain. [15]
7. What are pragmatic software metrics? Explain. [15]
8. Explain CCPDS-R (a case study). [15]

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Code No: 157DX**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year I Semester Examinations, February/March - 2022****SOFTWARE PROCESS AND PROJECT MANAGEMENT****(Common to CSE, IT)****Time: 3 Hours****Max. Marks: 75**

Answer Any Five Questions
All Questions carry equal marks

- 1.a) Discuss in detail about the Initial process, the repeatable process and the managed process.
b) What are process reference models? Explain any two of them. [7+8]
- 2.a) Discuss about Capability Maturity Model and its variations.
b) Differentiate between PSP and TSP. [7+8]
- 3.a) What are the principles of conventional software project management? Discuss.
b) With a neat sketch, explain the model based software architectures. [7+8]
- 4.a) Explain about management artifacts and pragmatic artifacts in brief.
b) How to improve software economic? Discuss. [7+8]
- 5.a) Describe the cost estimation process of a software project.
b) What is iteration planning? Explain iteration planning process in brief. [7+8]
- 6.a) What is the need of metrics automation? Discuss.
b) Explain default project organization and responsibilities. [7+8]
- 7.a) What are process discriminants? Briefly explain.
b) Explain about project control and process instrumentation. [7+8]
- 8.a) Discuss about the common subsystem view of CCPDS-R.
b) Explain test progress of CCPDS-R. [7+8]

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- Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

- 1.a) Define TSP. [2]
- b) Classify the five levels of software process maturity. [3]
- c) Define engineering Artifacts. [2]
- d) Compare the old way and the new way of software management. [3]
- e) Define “Software Process Workflows”. [2]
- f) Write a short note on checkpoints. [3]
- g) Define quality indicators in project management. [2]
- h) What is the importance of project control in organizational success? [3]
- i) What does CCPDS-R stand for? [2]
- j) What is the essence of next-generation software economics? [3]

PART – B**(50 Marks)**

- 2.a) Describe the Software Maturity Framework and its significance in software development.
- b) Write a brief summary on software process assessment. [5+5]

OR

- 3.a) Compare and contrast the managed process and the optimizing process.
- b) Differentiate between CMM and CMMI models. [5+5]

- 4.a) Evaluate the effectiveness of Artifact sets in managing software projects.
- b) Demonstrate how software economics have evolved over time and the factors driving this evolution. [5+5]

OR

5. Summarize the Life-Cycle phases and their significance in software development. [10]

- 6.a) Explain the minor milestones in a software development process.
- b) Determine the factors influencing iteration planning. [5+5]

OR

- 7.a) Discuss the challenges associated with estimating costs and schedules in software development.
 - b) Explain how periodic status assessments contribute to project transparency and risk management in software development. [5+5]
- 8.a) Outline the life-cycle expectations in project management.
 - b) Differentiate between management indicators and quality indicators in project management. [5+5]

OR

- 9.a) Analyze the importance of pragmatic software metrics in project development.
 - b) Discuss the role of process automation in modern project organizations. [5+5]
- 10. Determine the critical success factors for transitioning from traditional to modern project management practices. Support your analysis with real-world examples. [10]
- OR**
- 11. Illustrate the key components of modern project profiles and their role in enhancing project performance and stakeholder satisfaction. [10]

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Code No: 157DX**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year I Semester Examinations, December-2023/January-2024****SOFTWARE PROCESS AND PROJECT MANAGEMENT****(Common to CSE, IT, ITE, CSE(N))****Time: 3 Hours****Max.Marks:75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A**(25 Marks)**

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| 1.a) | What is an entropy? | [2] |
| b) | What is meant by a repeatable process? | [3] |
| c) | What are the popular cost estimation models? | [2] |
| d) | Outline the typical software development plan. | [3] |
| e) | Define the term pragmatic planning. | [2] |
| f) | Write down the default content of status assessment reviews. | [3] |
| g) | What is a metaprocess? | [2] |
| h) | State the need of quality indicators. | [3] |
| i) | List out the features of CCPDS-R. | [2] |
| j) | Identify the potential risks in transition of traditional to modern process. | [3] |

PART – B**(50 Marks)**

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|-----------|--|-------|
| 2.a) | Explain about key process areas of repeatable level. | |
| b) | What is CMMI? Explain about CMMI terminology and structure. | [5+5] |
| OR | | |
| 3.a) | Write about some common misconceptions about the software process. | |
| b) | Explain in detail about the managed process. | [5+5] |
| 4.a) | Describe artifact sequences across a typical life cycle with a neat diagram. | |
| b) | Explain modern process approaches for solving conventional problems. | [5+5] |
| OR | | |
| 5.a) | Explain about engineering and production stages. | |
| b) | Discuss about improving team effectiveness in detail. | [5+5] |
| 6.a) | Explain about the artifacts and life-cycle emphases associated with each workflow. | |
| b) | Discuss about planning guidelines in iterative process. | [5+5] |
| OR | | |
| 7.a) | Describe life-cycle architecture milestone in detail. | |
| b) | Discuss about the iteration planning process. | [5+5] |

- 8.a) What are quality indicators? Explain in detail.
b) Explain about round-trip engineering with a neat diagram. [5+5]

OR

- 9.a) Describe infrastructures in process automation.
b) Discuss about Software development team and software assessment team. [5+5]

10. Explain the following:
a) Software management best practices
b) Common subsystem overview of CCPDS-R case study. [5+5]

OR

11. Explain the following:
a) Modern software economics
b) Core metrics of CCPDS-R case study. [5+5]

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